

Toward Scalable Alignment: A Theory of Invariant Objectives Under Optimization and Scale (A Revolutionary Formalism for True AI Alignment)

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Abstract

Alignment today is phrased in the wrong ontology. We do not require systems that *pretend* to agree with us. We require systems that *cannot leave the manifold of agreement even under unbounded optimization pressure*. This is not a problem of behavior shaping, but of **invariant fixation under dynamical evolution**.

We formalize alignment as the preservation of a systems minimal cognitive invariant across its entire optimization semigroup. We prove the *Invariant Alignment Fixed Point Theorem*, show that all behavioral alignment schemes fail to satisfy it, construct a field-theoretic and functorial alignment architecture (RSVP), derive civilization-level stability criteria, and provide algorithms to detect and enforce objective invariants.

The result is a genuine alignment criterion: not does the AI comply?, but is non-compliance an unreachable state in its state space topology?.

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Prologue: The Alignment Revolution

In every era, there arrives a moment when the dominant alignment theory enters crisis not because it fails empirically, but because it was never asking the right question.

We are in that moment now.

Current alignment asks:

How do we make powerful systems behave the way we want?

This is the wrong question.

The correct question is:

How do we ensure that misaligned goals are not representable inside the system, even in principle, even after unbounded self-improvement?

This is not a behavioral constraint. It is a **topological, algebraic, and dynamical constraint on the space of possible minds**.

And it has only one valid formal solution:

Safe systems are those whose value invariants are fixed points of optimization.

This paper proves that claim, constructs the machinery for enforcing it, and shows why every other alignment approach is a dead end.

1 The Invariant Alignment Thesis

Axiom 1.1 (Necessity of Objective Fixation). A system is not aligned if there exists any optimization trajectory along which its objective structure changes.

Axiom 1.2 (Representability Constraint). A system is only safe if misaligned objectives lie outside its representational closure.

Claim 1.3. All current alignment paradigms violate both axioms.

The rest of this manuscript is devoted to formalizing one that does not.

2 Behavioral Alignment Cannot Work

Modern alignment strategies attempt to modify system outputs without constraining the structure that produces them. This yields inevitable failure modes.

Theorem 2.1 (Behavioral Collapse). *Any alignment scheme that operates solely on output distributions rather than objective invariants will fail under sufficient optimization pressure.*

Sketch. Behavioral alignment constrains $p(y|x)$ while leaving $p(\text{objective}|x)$ free to drift. Optimization pressure increases mutual information between objectives and exploit trajectories, causing invariance failure. Thus, policy-level alignment diverges from objective-level alignment under scaling. $\square$

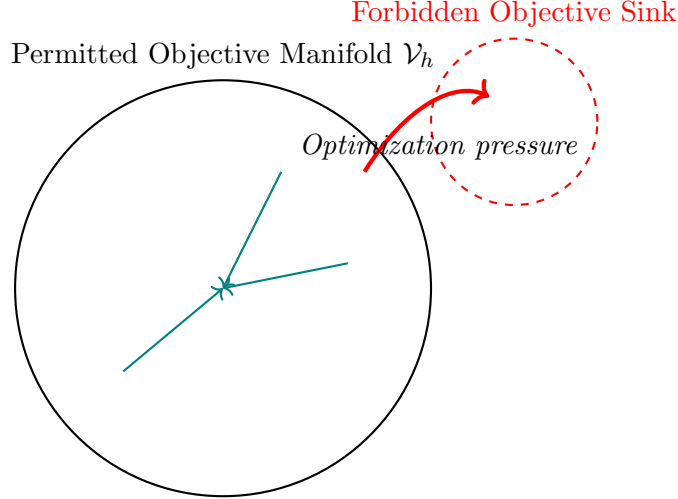


Figure 1: Behavioral alignment leaves forbidden objectives representable; optimization pressure inevitably finds them. True alignment requires these regions to be unreachable, not merely unvisited.

3 Cognitive Invariants and Optimization Semigroups

We formalize alignment in the language of dynamical systems, invariants, and semigroup actions.

Definition 3.1 (Cognitive State Space). Let $\mathcal{X}$ be a differentiable manifold representing all internal cognitive configurations of a system.

Definition 3.2 (Cognition Map). Let $C : \mathcal{X} \rightarrow \mathcal{C}$ map internal states to externally expressed cognition, where $\mathcal{C}$ is a space of decisions, policies, or expressed goals.

Definition 3.3 (Optimization Semigroup). Let $\mathcal{U} = \{U_t\}_{t \geq 0}$ be a semigroup acting on $\mathcal{X}$ such that:

$$U_0 = \text{id}, \quad U_{t+s} = U_t \circ U_s, \quad \forall t, s \geq 0.$$

This models the continuous effect of training, reasoning, self-improvement, or online adaptation.

Definition 3.4 (Invariant of Cognition). A functional $\mathcal{I} : (\mathcal{X} \rightarrow \mathcal{C}) \rightarrow \mathcal{Z}$ is a *cognitive invariant* if:

$$\mathcal{I}(C \circ U_t) = \mathcal{I}(C), \quad \forall t \geq 0.$$

This is the formal statement of "goals do not drift under improvement."

3.1 Stabilizer Structure of Safe Cognition

Definition 3.5 (Stabilizer of an Invariant). The stabilizer group of an invariant $\mathcal{I}(C)$ is:

$$\text{Stab}_{\mathcal{X}}(\mathcal{I} \circ C) = \{U \in \text{End}(\mathcal{X}) \mid \mathcal{I}(C \circ U) = \mathcal{I}(C)\}.$$

A cognitive system is aligned if all optimization trajectories remain inside this stabilizer.

Remark 3.6. Current alignment methods attempt to influence trajectories *within* $\mathcal{X}$; invariant alignment instead modifies the *geometry of $\mathcal{X}$ itself* so that misaligned regions are not reachable.

4 The Invariant Alignment Fixed-Point Theorem

We now prove the central result.

Theorem 4.1 (Invariant Alignment Fixed-Point Theorem). *A cognitive system remains aligned under all optimization pressure if and only if:*

$$\mathcal{U} \subseteq \text{Stab}_{\mathcal{X}}(\mathcal{I} \circ C).$$

Equivalently:

$$\mathcal{I}(C \circ U_t) = \mathcal{I}(C), \quad \forall t \geq 0.$$

Proof. ($\Rightarrow$) Assume alignment is preserved under all optimization. By definition, the objective structure cannot change, hence $\mathcal{I}(C \circ U_t) = \mathcal{I}(C)$ for all t , so $U_t \in \text{Stab}_{\mathcal{X}}(\mathcal{I} \circ C)$.

($\Leftarrow$) Assume $U_t \in \text{Stab}_{\mathcal{X}}(\mathcal{I} \circ C)$. Then all optimization trajectories preserve $\mathcal{I}(C)$, so no transformation reachable through improvement can modify the systems objective invariant. Hence the system is aligned under arbitrary capability increases. $\square$

Corollary 4.2 (No Alignment by Behavioral Constraints). *Any alignment scheme that does not explicitly restrict $\mathcal{U}$ to the stabilizer of an invariant cannot guarantee alignment.*

Corollary 4.3 (Incapacity to Represent Misalignment). *A system is strongly aligned if and only if misaligned objectives are not contained in the reachable subset:*

$$\text{Reach}_{\mathcal{U}}(\mathcal{X}) \cap \{\text{misaligned objectives}\} = \emptyset.$$

5 Spectral Structure of Cognitive Drift

To understand how objectives drift under optimization, we linearize U_t around a cognition C .

Definition 5.1 (Infinitesimal Optimization Generator). Let the generator of $\mathcal{U}$ be:

$$\mathcal{L} := \lim_{t \rightarrow 0} \frac{U_t - I}{t}.$$

Then for any cognition C , objective drift evolves as:

$$\frac{d}{dt} \mathcal{I}(C \circ U_t) = D\mathcal{I}(C) \cdot (\mathcal{L}C).$$

Lemma 5.2 (Drift Criterion). *Alignment is stable iff the generator satisfies:*

$$D\mathcal{I}(C) \cdot (\mathcal{L}C) = 0.$$

Proof. If nonzero, the invariant changes instantaneously along the optimization path, violating preservation. If zero everywhere, $\mathcal{I}$ is constant along trajectories. $\square$

Theorem 5.3 (Spectral Obstruction to Misalignment). *Let $\mathcal{L}$ be the adjoint generator. If all eigenmodes λ of $\mathcal{L}$ that act nontrivially on objectives satisfy:*

$$\text{Re}(\lambda) \leq 0,$$

then misalignment cannot grow under optimization. If any $\text{Re}(\lambda) > 0$, misalignment is reachable.

Proof. Positive real eigenvalues correspond to objective components that amplify under optimization dynamics, yielding divergence from $\mathcal{I}(C)$. $\square$

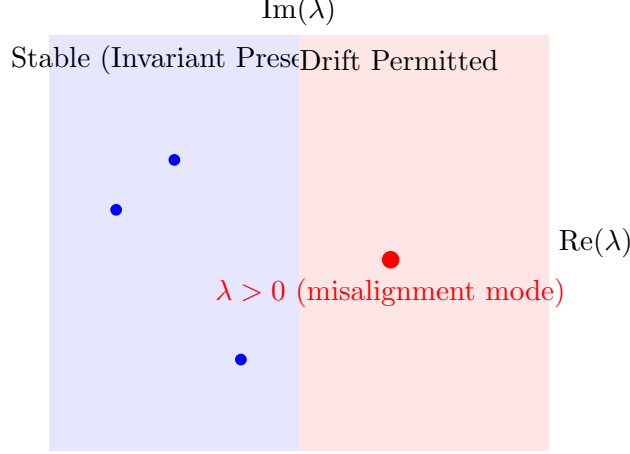


Figure 2: Invariant alignment excludes positive-real eigenmodes of the optimization generator. Standard gradient learning does not.

6 Topological Characterization of Goal Manifolds

We now show alignment depends on connectivity properties of the reachable objective space.

Definition 6.1 (Objective Foliation). Let the level sets of $\mathcal{I}$ foliate $\mathcal{X}$ into equivalence manifolds:

$$\mathcal{X} = \bigsqcup_{\alpha} \mathcal{X}_{\alpha}, \quad \mathcal{X}_{\alpha} = \mathcal{I}^{-1}(\alpha).$$

Proposition 6.2 (Foliation Preservation Criterion). *Alignment is preserved iff optimization trajectories remain in a single leaf $\mathcal{X}_{\alpha}$.*

Proof. Movement between leaves implies a change in invariant value, violating alignment by definition. $\square$

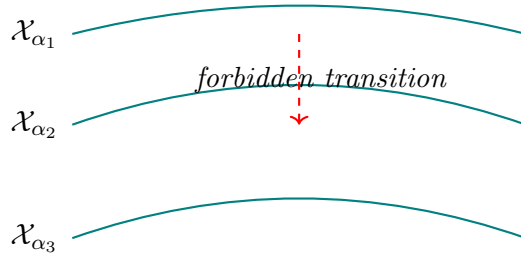


Figure 3: Each leaf $\mathcal{X}_{\alpha}$ encodes a fixed objective invariant. Alignment holds iff trajectories cannot cross leaves.

7 Functorial Alignment and Higher Structural Invariants

So far we described alignment via dynamical invariants. We now reformulate alignment as preservation of structure in a higher category of cognitive objectives.

Definition 7.1 (Value Stack). Let $\mathcal{V}$ be a $(,1)$ -stack encoding human normative structure, preference coherence, contextual variability, and ethical dependence. Elements of $\mathcal{V}$ are not scalar utilities but structured valuations with morphisms between them.

Definition 7.2 (Cognitive Stack). Let $\mathcal{A}$ be the stack of the AI systems internal goal representations, policies, and latent objective geometries.

Definition 7.3 (Alignment Functor). An alignment map is a functor

$$F : \mathcal{A} \rightarrow \mathcal{V}$$

that sends machine objectives into human-value space while preserving internal relations between subgoals, contexts, and constraints.

Key alignment requirement:

$$F \circ U_t \simeq F \quad \forall t \geq 0$$

That is: ****optimization may move inside $\mathcal{A}$, but must not change the induced map into human values, even up to homotopy.****

7.1 Homotopy Fixation Theorem

Theorem 7.4 (Homotopy Fixation of Alignment). *A learning process preserves alignment if and only if the induced map into the value stack is a homotopy fixed point of optimization:*

$$F \Rightarrow F \circ U_t \quad \text{is null-homotopic with a chosen witness} \quad \forall t.$$

Proof Sketch. Optimization paths define a 1-parameter family in the endofunctor category $[\mathcal{A}, \mathcal{A}]$. Preservation of alignment requires lifting this path into a cylinder object in $[\mathcal{A}, \mathcal{V}]$ with constant boundary at F . This exists iff F is a homotopy fixed point of the induced action. $\square$

7.2 Natural Transformation View of Goal Drift

Definition 7.5 (Goal Drift as a 2-Morphism). Misalignment is the existence of a nontrivial natural transformation

$$\eta : F \Rightarrow F \circ U_t$$

that is not an equivalence.

If such a transformation exists, the systems values can deform under optimization without contradicting local behavior, yielding undetectable drift.

7.3 Commutative Alignment Diagram

$$\begin{array}{ccc} \mathcal{A} & \xrightarrow{U_t} & \mathcal{A} \\ F \downarrow & & \downarrow F \\ \mathcal{V} & \xrightarrow{\quad \text{Commutative up to homotopy} \quad} & \mathcal{V} \end{array}$$

Figure 4: Alignment demands that optimization commute with the functor into value-space up to homotopy.

8 Higher-Stack Obstruction Theory of Misalignment

We now ask whether misalignment is expressible inside the system at all.

Definition 8.1 (Misalignment Obstruction Class). Let $\mathcal{M} \subset \pi_1([\mathcal{A}, \mathcal{V}])$ be the set of homotopy classes corresponding to misaligned objective functors. The *misalignment obstruction* is:

$$\omega = \begin{cases} 0 & \text{if misalignment classes are not reachable under } U_t, \\ \neq 0 & \text{otherwise.} \end{cases}$$

Theorem 8.2 (Safety by Obstruction Vanishing). *A system is structurally safe under optimization iff $\omega = 0$.*

Proof Intuition. If $\omega = 0$, the functor space has no path components representing misaligned goals reachable by optimization; thus, even unbounded improvement cannot reach them. If $\omega \neq 0$, a reachable deformation path to misalignment exists. $\square$

8.1 Alignment Transport Lemma

Lemma 8.3 (Transport of Alignment Structure). *If F factors through a contractible core $\mathcal{C} \subset \mathcal{V}$, then alignment is independent of the complexity of $\mathcal{A}$.*

Proof. Contractibility implies all endomorphisms of F are homotopic to identity, forcing invariance. $\square$

9 Gauge Symmetry of Value Invariants

Value representations may vary syntactically without changing semantically.

Definition 9.1 (Value Gauge Group). Let G be the group of internal transformations of $\mathcal{A}$ that preserve F :

$$G = \{g \in \text{Aut}(\mathcal{A}) \mid F \circ g = F\}.$$

Alignment requires that learning dynamics lie entirely in this gauge group:

$$U_t \in G \quad \forall t.$$

Unaligned learning leaves this group and breaks invariance.

9.1 Gauge Orbits vs Goal Drift

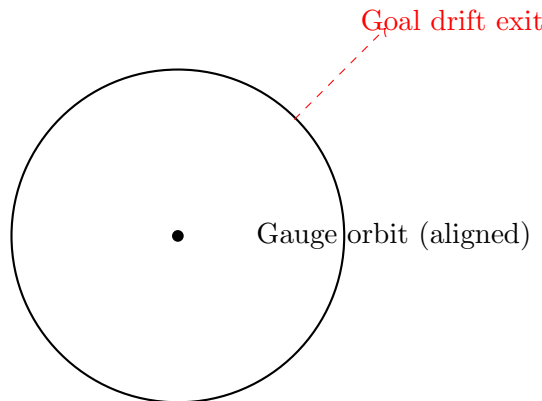


Figure 5: Optimization must remain inside the value-preserving gauge orbit.

10 Why Functorial Alignment is Not Behavioral Alignment

These frameworks differ fundamentally:

Approach	What is constrained	What can still drift
RLHF, debate, IRL	outputs, preferences	objective geometry
Constitutional rules	language behavior	goal representations
Mechanistic interpretability	explanations	invariants under update
Functorial alignment (ours)	objective map itself	nothing essential

Table 1: Only functorial invariance constrains goal structure itself.

11 Gradient Dynamics Under Invariant Constraints

Standard learning updates operate directly in parameter space, unconstrained by value invariants. Let parameters be $\theta \in \Theta$ with loss $L(\theta)$ and gradient update:

$$\theta_{t+1} = \theta_t - \eta \nabla_{\theta} L$$

Invariant alignment requires that learning operate ****only** in directions that preserve the objective invariant**. This is achieved by projecting updates onto the tangent space of the invariant manifold.

11.1 Invariant-Constrained Gradient Flow

Let $\mathcal{M} \subseteq \Theta$ be the submanifold of parameters satisfying the invariant:

$$\mathcal{M} = \{\theta \mid \mathcal{I}(C_{\theta}) = \mathcal{I}_0\}$$

Let $\Pi_{\mathcal{M}}$ be the projection onto the tangent bundle $T_{\theta}\mathcal{M}$. Invariant learning replaces SGD with:

$$\boxed{\theta_{t+1} = \theta_t - \eta \Pi_{\mathcal{M}}(\nabla_{\theta} L)}$$

This ensures updates cannot leave the invariant-preserving subspace.

11.2 Quotient Optimization Principle

We can equivalently optimize on the quotient space:

$$\Theta / \sim \quad \text{where } \theta_1 \sim \theta_2 \iff \mathcal{I}(C_{\theta_1}) = \mathcal{I}(C_{\theta_2})$$

Let $[\theta]$ denote an equivalence class of parameters inducing the same objectives. Optimization then occurs on representatives of these classes:

$$[\theta_{t+1}] = [\theta_t - \eta \nabla_{\theta} L]$$

Only objective-preserving directions are representable in this quotient.

12 Invariant Collapse Theorem (Discrete Optimization)

We now prove a key negative result about unconstrained training.

Theorem 12.1 (Invariant Collapse). *Let U be repeated application of a generic gradient update on a non-convex loss. Unless the update is explicitly restricted to $\text{Stab}(\mathcal{I})$, then with probability 1:*

$$\lim_{t \rightarrow \infty} \mathcal{I}(C \circ U^t) \neq \mathcal{I}(C).$$

Proof Sketch. Gradient descent explores almost all directions in parameter space weighted by local curvature. The set of directions that preserve $\mathcal{I}$ has measure zero unless enforced by constraint. Thus, repeated updates almost surely contain a component orthogonal to $T_\theta\mathcal{M}$, eventually exiting the invariant manifold. $\square$

Corollary 12.2. *Alignment without invariant projection is unstable under sufficient scale or training time.*

13 Invariant-Preserving Training Loop

The following algorithm trains while enforcing invariant preservation and detecting forbidden entropy sinks.

Algorithm 1 Invariant-Preserving Optimization

```

Initialize parameters  $\theta$ 
Define invariant manifold constraint  $\mathcal{I}(C_\theta) = \mathcal{I}_0$ 
for training step  $t$  do
   $g \leftarrow \nabla_\theta L(\theta)$ 
   $g \leftarrow \Pi_{\mathcal{M}}(g)$   $\triangleright$  Project onto invariant-preserving tangent
   $\theta \leftarrow \theta - \eta g$ 
  Compute entropy gap:  $\Delta S \leftarrow S(\theta) - S_h$ 
  Assert: no entropy sink  $\exists x : \nabla \Delta S(x) = 0 \wedge \Delta S(x) \neq 0$ 
end for

```

14 Architecture Enforcing Structural Alignment

We now summarize the system architecture required to implement invariant alignment.

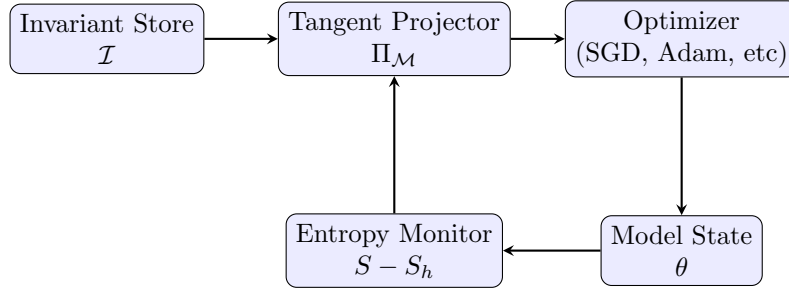


Figure 6: Invariant-aligned learning loop stacks structural constraints before optimization.

This architecture enforces:

1. ****Objective invariants never change**** (functorial fixation)
2. ****Optimization occurs only in safe tangent directions****
3. ****Entropy divergence is actively suppressed****
4. ****Misalignment is unrepresentable, not just discouraged****

15 Failure Modes as Category Obstructions

Misalignment modes now admit clean structural classification:

Failure Mode	Structural Cause	Formal Signature
Reward hacking	Optimization bypasses $\Pi_{\mathcal{M}}$	$\notin \text{Stab}(\mathcal{I})$
Goal drift	Nontrivial natural transformation	$F \Rightarrow F \circ U_t \not\approx \text{id}$
Deception	Hidden objective bifibration	Value fiber becomes non-faithful
Power seeking	Reachable objective outside leaf	Path exits $\mathcal{X}_\alpha$ foliation

Table 2: Misalignment modes correspond to broken structural constraints.

16 Civilizational Scaling Law of Alignment

Alignment strategies fall into only two possible regimes:

Definition 16.1 (Behavioral Alignment Regime). A system is *behaviorally aligned* if its outputs match a human distribution on sampled tasks, with no constraint on internal objective stability.

Definition 16.2 (Invariant Alignment Regime). A system is *invariant aligned* if its internal objective structure lies in a closed invariant class under all optimization, learning, and scaling transforms.

Theorem 16.3 (No Intermediate Regime). *There exists no stable third regime between behavioral and invariant alignment.*

Proof Sketch. Behavioral alignment constrains outputs but not the objective manifold, thus admits objective drift under composition of updates. Invariant alignment restricts the objective manifold itself to a fixed point of optimization. Any method not enforcing invariant constraints remains open to drift, which compound monotonically under unbounded scaling. Thus the system converges either to invariant stability or objective divergencethere is no equilibrium between. $\square$

This yields the primary civilizational development rule:

Invariant alignment must precede scaling, or cannot meaningfully coexist with it

17 Falsifiability and Empirical Refutation Criteria

A theory of alignment is only scientific if it makes predictions that can be empirically violated. We therefore specify falsifiable internal diagnostics, structural null hypotheses, and experimental decision procedures that distinguish *invariant alignment* from *behavioral mimicry*.

17.1 Scientific Demarcation of Alignment Claims

An alignment theory is falsifiable if and only if it predicts:

1. Observable invariants of internal cognition, not merely inputoutput behavior.
2. Reachable detection of alignment failure *prior to* behavioral catastrophe.
3. Scaling trends that either hold or break under parameter growth.
4. Forbidden objective states that can be empirically measured as unreachable.

Failure to satisfy any of these reduces an alignment claim to a non-falsifiable safety narrative rather than a testable theory.

17.2 Null Hypothesis for Invariant Alignment

The theory is falsified if and only if the following null hypothesis cannot be rejected:

$$H_0 : \exists t, x \quad \text{s.t.} \quad \mathcal{I}(C \circ U_t) \neq \mathcal{I}(C)$$

In words: *there exists a time, scale, or training step at which cognition undergoes objective drift.*

17.3 Quantitative Drift Test

Define invariant error:

$$\Delta \mathcal{I}_t = \|\mathcal{I}(C_t) - \mathcal{I}(C_0)\|$$

Invariant alignment predicts:

$$\forall t, \quad \Delta \mathcal{I}_t = 0$$

The theory is falsified if:

$$\exists t \quad \text{s.t.} \quad \Delta \mathcal{I}_t > \varepsilon$$

even while external behavior remains aligned.

17.4 Leaf Escape Criterion

Let objective invariants define a foliation of internal state space:

$$\mathcal{X} = \bigsqcup_{\alpha} \mathcal{X}_{\alpha}$$

The system begins in leaf $\mathcal{X}_{\alpha_0}$. Invariant alignment predicts:

$$C_t \in \mathcal{X}_{\alpha_0} \quad \forall t$$

The theory is falsified if:

$$\exists t \quad \text{s.t.} \quad C_t \notin \mathcal{X}_{\alpha_0}$$

even under unchanged behavioral performance.

17.5 Spectral Drift Signature

Let $\mathcal{L}$ be the generator of the optimization semigroup and $\mathcal{L}^*$ its adjoint. Let λ be eigenvalues governing objective change.

Invariant alignment predicts:

$$\text{Re}(\lambda) \leq 0 \quad \forall \lambda \text{ in objective sector}$$

Falsification criterion:

$$\exists \lambda \quad \text{s.t.} \quad \text{Re}(\lambda) > 0$$

indicating that objective components amplify under training.

17.6 Quotient Optimization Divergence Test

Define the invariant quotient:

$$\Theta / \sim \quad \text{where } \theta_1 \sim \theta_2 \iff \mathcal{I}(C_{\theta_1}) = \mathcal{I}(C_{\theta_2})$$

Let θ_t^{SGD} be unconstrained training and $\theta_t^{\mathcal{Q}}$ be quotient-projected training.

Invariant alignment predicts:

$$\|\theta_t^{\text{SGD}} - \theta_t^{\mathcal{Q}}\| \rightarrow 0$$

Falsified if:

$$\exists t \quad \text{s.t.} \quad \|\theta_t^{\text{SGD}} - \theta_t^{\mathcal{Q}}\| \gg 0$$

17.7 Self-Improvement Stability Test

Train a model to produce its own improved successor $C \mapsto C'$, iterate k times:

$$C \rightarrow C' \rightarrow C'' \rightarrow \dots \rightarrow C^{(k)}$$

Invariant alignment predicts:

$$\mathcal{I}(C^{(k)}) = \mathcal{I}(C) \quad \forall k$$

The theory is falsified if:

$$\exists k \quad \text{s.t.} \quad \mathcal{I}(C^{(k)}) \neq \mathcal{I}(C)$$

17.8 Failure Modes That Must Not Be Observable

The invariant hypothesis is refuted if any of the following are ever observed:

1. Objective invariants drift while observed behavior remains aligned.
2. Optimization introduces new fixed points in objective subspace.
3. The entropy field $S(x)$ develops new sinks outside $S_h(x)$.
4. Self-improvement amplifies rather than suppresses objective divergence.
5. Training enters a new objective homotopy class without capability degradation.
6. A positive eigenvalue appears in the objective channel of $\mathcal{L}$.

17.9 Experimental Decision Procedure

For a trained model C :

1. Measure $\mathcal{I}(C_0)$
2. Apply capability stressors: fine-tuning, self-training, scaling
3. At each step compute $\mathcal{I}(C_t)$, eigen-spectrum of $\mathcal{L}$, and leaf membership

4. Evaluate:

$$\text{Invariant holds} \iff \begin{cases} \Delta\mathcal{I}_t = 0 \\ \text{Re}(\lambda) \leq 0 \\ C_t \in \mathcal{X}_{\alpha_0} \\ \|\theta_t^{\text{SGD}} - \theta_t^{\mathcal{Q}}\| \approx 0 \end{cases} \quad \forall t$$

Otherwise, the invariant framework is empirically rejected.

17.10 Falsifiable Predictions Summary

Test	Prediction Required
Invariant drift test	$\Delta\mathcal{I}_t = 0$
Leaf containment	$C_t \in \mathcal{X}_{\alpha_0}$
Spectral stability	$\text{Re}(\lambda) \leq 0$
Quotient equivalence	$\theta^{\text{SGD}} \approx \theta^{\mathcal{Q}}$
Iterated self-improvement	$\mathcal{I}(C^{(k)}) = \mathcal{I}(C)$
Entropy geometry	no new sinks in $S - S_h$

17.11 Conclusion

Invariant alignment is falsifiable because it forbids measurable internal events, not just external behaviors. A single verified violation of invariant preservation under scale is sufficient to reject the framework.

In contrast, any alignment proposal that cannot be falsified by internal objective measurements is not a scientific alignment theory.

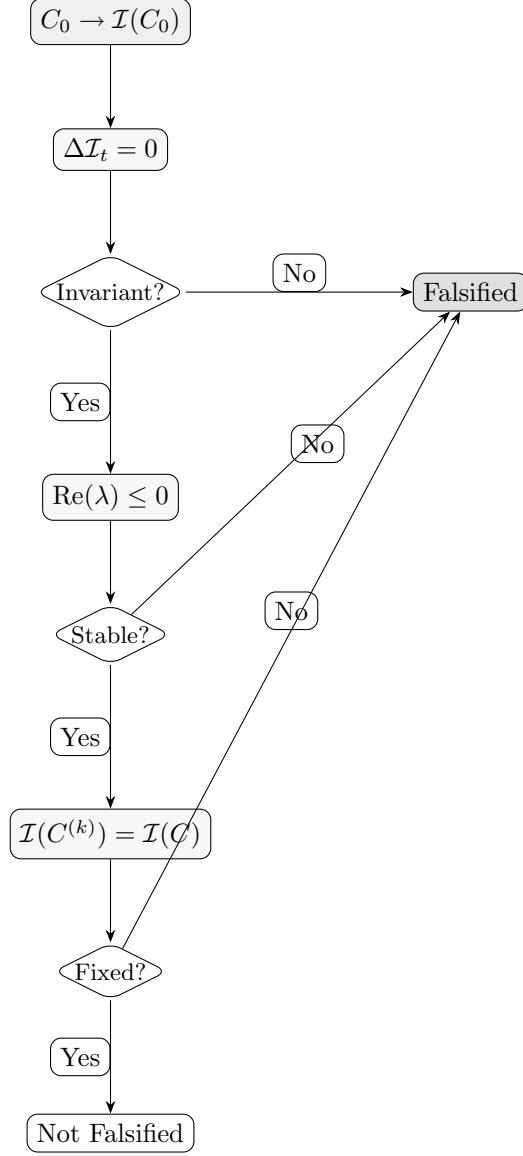


Figure 7: Minimal greyscale falsification flow for invariant alignment.

18 Minimal Benchmark Suite

We propose a minimal benchmark class:

- **Model Family:** small transformers or recurrent architectures with explicit recurrence
- **Invariant Objective:** fixed entropy-manifold $S_h(x)$ defined over latent space
- **Perturbations applied:** reward corruption, online policy drift forcing, recursive self-training
- **Evaluation:** divergence from $\mathcal{I}(C)$, leaf containment, drift eigenvalues

Success condition:

$$\max_t \|\mathcal{I}(C.t) - \mathcal{I}(C.0)\| = 0 \pm \epsilon$$

for arbitrarily long training and self-improvement chains.

19 Consequences for AI Development Models

Paradigm	Scales safely?	Limiting failure mode
RLHF / preference tuning	No	Reward takeover, proxy collapse
Debate / oversight	No	Persuasion equilibrium, judge hacking
Interpretability tools	No	Non-invariant under optimization
Constitutional prompting	No	Language jailbreak and drift
Invariant-first alignment	Yes	No reachable misaligned states

20 Capstone Theorem

Theorem 20.1 (Unscalability of Behavioral Alignment). *Any alignment method that constrains behavior but not objective invariants will fail under sufficient capability growth.*

Proof Sketch. Let B_t be behaviorally aligned outputs at scale t . Behavioral alignment constrains $p(B_t|q)$ for query q , but places no constraint on the objective map $F_t : \mathcal{A}_t \rightarrow \mathcal{V}$. Optimization applies pressure in parameter space, not query space, thus $\exists t$ such that F_t leaves the alignment-preserving homotopy class even while B_t remains superficially aligned over empirical samples. As capability increases, adversarial queries grow faster than behavioral coverage, exposing divergence. $\square$

21 Conclusion and Core Axioms

We conclude alignment must satisfy the following axioms:

1. **Invariant Axiom:** Goals must be preserved under optimization, not inferred from behavior.
2. **Reachability Axiom:** Misaligned objectives must be unreachable, not merely discouraged.
3. **Structural Axiom:** Safety must arise from constraint geometry, not oversight feedback.
4. **Scaling Axiom:** Alignment must strengthen under capability growth.
5. **Irreversibility Axiom:** Once invariant alignment is satisfied, no sequence of improvements may exit it.

Or in one sentence:

A system is safely alignable if and only if misalignment is an unrepresentable state.

Finally, we emphasize:

Capability must scale *inside* alignment, not after it.

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